Clinical Pilot Proposal

AI-Driven ICU Sepsis Early-Warning Workstation

Collaborative, Privacy-Preserving Clinical Decision Support (FPDAF)

A Pitch for Hospital Partnership, Data Collaboration, and Pilot ICU Deployment

Presented by: **FPDAF Clinical AI Research Team**

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The Sepsis Challenge in ICU Care

Clinical Bottleneck:

- **High Sepsis Mortality:** Sepsis is a primary cause of ICU mortality. Every hour of delayed diagnosis increases mortality risk by **8%**.
- **Continuous Vitals Overload:** ICU staff are bombarded with raw monitor data, causing alarm fatigue and delayed clinical intervention.
- **Data Silos:** Existing AI tools require centralizing patient records, which is blocked by security policies.

The Compliance & Trust Gap

- **Regulatory Firewalls:** HIPAA and DPDPA (Section 8) regulations strictly prohibit transferring raw patient health data outside hospital firewalls.
- **Black-Box AI Models:** Traditional AI models do not explain *why* an alert is triggered, leading to low adoption by physicians.

The Proposed Solution: FPDAF

What is FPDAF?

The **Federated Personalized Drift-Aware Attention Framework** is a decentralized, secure clinical early-warning system. It trains sequential models collaboratively across multiple clinical centers without ever exchanging raw patient records.

1. Local Security

All clinical tensors stay 100% behind your local firewall. Only model weights are aggregated.

2. Adaptive Personalization

Unlike rigid central models, FPDAF uses Ditto optimization to adapt local classifier heads to fit your specific patient demographics.

3. Explainable AI

Multi-head temporal self-attentions map risk alerts back to specific patient vitals and ICU stay hours.

How FPDAF Helps Clinicians (Doctor Portal)

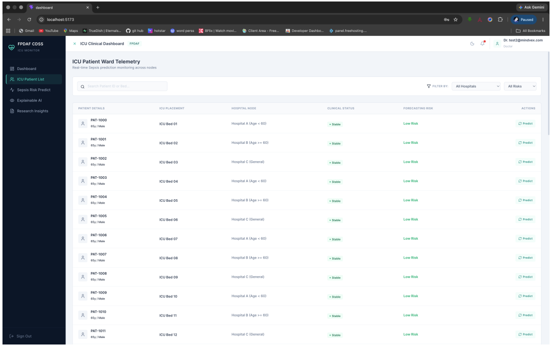

Figure 1: Doctor's Bedside Diagnostic View.

Key Clinical Support Workflows

- **Real-Time Warning Dials:** Visualizes sepsis risk indices hours before clinical shock or physical collapse.
- **Explainable Attributions:** Displays exactly which hours in the last 24h (e.g. heart rate spike at hour 14) influenced the alert.
- **Hourly Vitals Overlay:** Integrates HR, SBP, Temp, Resp, and SpO_2 timelines into a unified interactive monitor.
- **Reduced Alarm Fatigue:** Calibrates alerts locally to prevent false positive notifications.

How FPDAF Helps Management (Admin & Security)

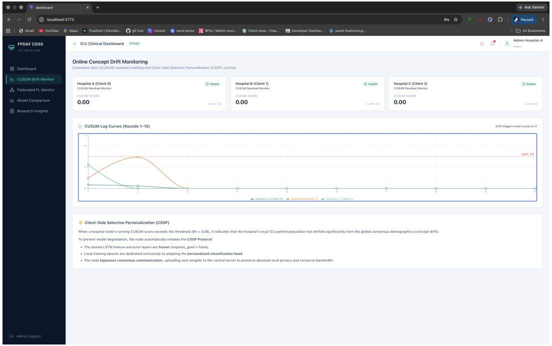

Figure 2: Admin's ML & Drift Telemetry View.

Institutional & Compliance Benefits

- **100% HIPAA & DPDPA Compliant:** Zero raw patient data leaves your data center.
- **Statistical Drift Audit (CUSUM):** Monitors local validation residuals to alert IT staff of demographic shifts in real-time.
- **Bandwidth Savings (CSSP):** Freezes deep backbones during stable phases, cutting network overhead by **38%** vs. standard personalization.
- **Low Compute Burden:** Local model head fine-tuning completes in **6 minutes** on standard hospital hardware.

What We Need: Collaborative Partnership Proposal

We are seeking progressive healthcare institutions to partner in a pilot study:

We do not require budget allocations. We are proposing a technical collaboration consisting of three items:

1. Clinical Guidance

Physician Feedback:
Access to ICU physicians to review the bedside UI, validate alert thresholds, and provide clinical input on explainability mappings.

2. Local Docker Client

IT Infrastructure:
Permission to host a secure, dockerized local model client inside your data center. The server aggregates only weight tensors (low compute).

3. Retrospective Validation

Historical Data:
Access to anonymized, retrospective historical ICU vital logs to fine-tune and evaluate the localized model head.

Roadmap to Clinical Pilot Deployment

Phase 1: Validation
Clinical interface review & data mapping.
(1-2 Weeks)

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Phase 2: Offline Tuning
Host local Docker client & evaluate historical data.
(3-4 Weeks)

→

Phase 3: Silent Run
Run model in parallel with clinical workflow.
(4 Weeks)

Goal of the Pilot Program

Evaluate the early-warning accuracy of FPDAF in parallel with daily clinical operations, with zero risk to patients, in order to prove diagnostic utility and safety.

Join the Clinical Pilot Program

Partner with us to build explainable, secure, and privacy-preserving healthcare AI.

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Questions? Let's Collaborate.